

NIK O. GJONBALAJ

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EDUCATION

Harvard University, Cambridge, MA

Ph.D. in Physics (M.A. in Physics Nov 2025)

Advisor: [Susanne F. Yelin](#)

August 2022 - Present

GPA: 4.00/4.00

Boston University, Boston, MA

B.A. (with honors) / M.A. in Physics

Advisors: [Anatoli Polkovnikov](#), [A. Liam Fitzpatrick](#)

September 2018 - May 2022

GPA: 4.00/4.00

SCIENTIFIC PAPERS

[6] N. O. Gjonbalaj, C. Kokail, S. F. Yelin, and S. Choi, *Unified Probe of Quantum Chaos and Ergodicity from Hamiltonian Learning*, arXiv:2603.04486 (2026).

[5] A. A. Geim, N. U. Köyliüoğlu, S. J. Evered, R. Sahay, S. H. Li, M. Xu, D. Bluvstein, [N. O. Gjonbalaj](#), N. Maskara, M. Kalinowski, T. Manovitz, R. Verresen, S. F. Yelin, J. Feldmeier, M. Greiner, V. Vuletic, and M. D. Lukin, *Engineering quantum criticality and dynamics on an analog-digital simulator*, arXiv:2602.18555 (2026).

[4] X. De Vriendt, J. Bringewatt, [N. O. Gjonbalaj](#), S. Ostermann, D. Vodola, J. Borregaard, M. Kühn, and S. F. Yelin, *A New Angle on Quantum Subspace Diagonalization for Quantum Chemistry*, arXiv:2602.11985 (2026).

[3] [N. O. Gjonbalaj](#), R. Sahay, and S. F. Yelin, *Shortcuts to Analog Preparation of Non-Equilibrium Quantum Lakes*, arXiv:2502.03518 (2025). ([accepted to Phys. Rev. Lett.](#))

[2] [N. O. Gjonbalaj](#), S. Ostermann, and S. F. Yelin, *Modifying cooperative decay via disorder in atom arrays*, Phys. Rev. A **109**, 013720 (2024).

[1] [N. O. Gjonbalaj](#), D. K. Campbell, and A. Polkovnikov, *Counterdiabatic driving in the classical β -Fermi-Pasta-Ulam-Tsingou chain*, Phys. Rev. E **106**, 014131 (2022).

HONORS AND AWARDS

[Harvard Physics Gertrude and Maurice Goldhaber Prize](#)

April 2026

[Harvard Quantum Initiative Generation Q Fellowship](#)

July 2023

Harvard Physics Wallace-Noyes Fellowship

2023 - 2026

BU CAS Departmental Physics Award

May 2022

[BU Provost Scholar](#)

July 2020

[Barry Goldwater Scholar](#)

March 2020

[National Merit Scholarship Finalist](#)

Spring 2018

RESEARCH PRESENTATIONS

Oral Presentations

- [QSim](#) Contributed Talk (< 20% **accepted**), *Amsterdam, NL* 13 Aug 2026
“[Unified Probe of Quantum Chaos and Ergodicity from Hamiltonian Learning](#)”
- [UMD QuICS-JQI Special Seminar](#) (**invited**), *College Park, MD* 4 Jun 2026
“[From Preparing Quantum Spin Lakes to Measuring Chaos with Hamiltonian Learning: Quantum Simulation Beyond the Ground State](#)” (video)
- [SimQuDyn](#) Prize Poster Talk (**popular vote winner**), *Munich, DE* 2 Apr 2026
“[Shortcuts to Analog Preparation of Non-Equilibrium Quantum Lakes](#)”

- [ITAMP](#) Seminar (**invited**), *Cambridge, MA* 11 Feb 2026
“A Unified Probe of Quantum Chaos and Ergodicity from Hamiltonian Learning”
- [CUA](#) Retreat Presentation (**invited**), *Plymouth, NH* 21 Jan 2026
“Quantifying Quantum Chaos and Ergodicity via Hamiltonian Learning”
- [APS DAMOP Annual Meeting](#) 2025, *Portland, OR* 17 Jun 2025
“Quantum Ergodicity as the Robustness of Hamiltonian Learning”
- BU CMT Seminar (**invited**), *Boston, MA* 9 Apr 2025
“Shortcuts to Analog Preparation of Non-Equilibrium Quantum Lakes”
- [CUA](#) 10-Minute Talk (**invited**), *Cambridge, MA* 1 Apr 2025
“Quantifying Quantum Ergodicity from Hamiltonian Learning”
- [Sels](#) Group Talk (**invited**), *New York, NY* 21 Feb 2025
“Shortcuts to Analog Preparation of Non-Equilibrium Quantum Lakes”
- [APS DAMOP Annual Meeting](#) 2024, *Fort Worth, TX* 5 Jun 2024
“Non-Equilibrium Quantum Lakes from Counter-Diabatic Driving”
- [APS DAMOP Annual Meeting](#) 2023, *Spokane, WA* 7 Jun 2023
“Searching for Localization in Subradiant Atomic Arrays”
- [APS March Meeting](#) 2022, *Chicago, IL* 18 Mar 2022
“Minimizing Losses in Classical Nonlinear Oscillators”
- 2021 AFOSR AMP and QIS Program, *Online* 18 Aug 2021
“Minimizing Losses in Classical Nonlinear Oscillators”
- [APS March Meeting](#) 2021, *Online* 19 Mar 2021
“Minimizing Losses in a Classical Nonlinear Oscillator”
- Boston University CMS Group Meeting, *Boston, MA* 10 Dec 2019
“Optimization of the L1 Trigger for Axion-Like Dark Matter”

Poster Presentations

- [GRS](#) and [GRC](#), Quantum Science, *Easton, MA* 25 - 31 Jul 2026
“Unified Probe of Quantum Chaos and Ergodicity from Hamiltonian Learning”
- [SimQuDyn](#) 2026, *Munich, DE* 30 Mar - 1 Apr 2026
“Shortcuts to Analog Preparation of Non-Equilibrium Quantum Lakes”
- [Niels Bohr Quantum Summer School](#) 2025, *Copenhagen, DK* 15 Aug 2025
“Quantum Ergodicity as the Robustness of Hamiltonian Learning”
- [QSim](#) 2025, *New York, NY* 5 Aug 2025
“Shortcuts to Analog Preparation of Non-Equilibrium Quantum Lakes”
- [APS DAMOP Annual Meeting](#) 2025, *Portland, OR* 18 Jun 2025
“Quantum Ergodicity as the Robustness of Hamiltonian Learning”
- [CUA](#) Retreat, *Plymouth, NH* 20 Jan 2025
“Non-Equilibrium Quantum Lakes from Counter-Diabatic Driving”
- [FOR 5522 PhD School on Nonergodic Quantum Dynamics](#), *Göttingen, DE* 24 Sep 2024
“Non-Equilibrium Quantum Lakes from Counter-Diabatic Driving”
- [GRS](#) and [GRC](#), Quantum Science, *Easton, MA* 27 Jul - 2 Aug 2024
“Non-Equilibrium Quantum Lakes from Counter-Diabatic Driving”

- HQI-RIKEN Workshop, *Tokyo, JP* 18 Jun 2024
“Non-Equilibrium Quantum Lakes from Counter-Diabatic Driving”
- [CUA](#) Retreat, *Plymouth, NH* 15 Jan 2024
“Accelerating and Suppressing Cooperative Decay via Disorder in Atom Arrays in Free Space”
- Harvard QSE PhD Retreat, *Boston, MA* 10 Sep 2023
“Accelerating and Suppressing Cooperative Decay via Disorder in Atom Arrays in Free Space”
- [GRS](#) and [GRC](#), Quantum Control of Light and Matter, *Newport, RI* 5-11 Aug 2023
“Accelerating and Suppressing Cooperative Decay via Disorder in Atom Arrays in Free Space”
- [BU-PRO REU](#) Final Poster Session, *Boston, MA* 6 Aug 2021
“Minimizing Losses in Classical Nonlinear Oscillators”

TEACHING EXPERIENCE

Harvard University

- Teaching Fellow for P285A: Modern Atomic and Optical Physics I Spring 2025
- 3 Time Lecturer for P285A: Modern Atomic and Optical Physics I Spring 2025
- Completed [Physics 302A: Teaching and Communicating Physics](#) Spring 2023

Boston University

- [Learning Assistant](#) (Undergraduate Teaching Assistant)
 - [PY452](#): Quantum Physics II Fall 2021
 - [PY451](#): Quantum Physics I Spring 2021, Spring 2022
 - [PY408](#): Intermediate Mechanics Fall 2020
 - [PY106](#): Physics II Spring 2020

Tutoring

- AP Calculus Tutor Spring 2021
- High School Physics Tutor Fall 2017 - Spring 2018

COMMUNITY, MENTORING, AND OUTREACH

Outreach

- Committee Member for [EngageCUA](#) Spring 2025 - Present

Mentoring

- Polaris Mentor for Harvard Physics Undergraduates Fall 2022 - Present
- [PRISM](#) Mentor for Boston University Physics Freshmen Fall 2020 - Spring 2022
- Student Ambassador for [BU-PRO REU](#) Summer 2021

Journal Referee

- Quantum